The term metioned in the board volt is the measurement for the electric potential difference.

The SI unit of potential difference is volt (V). potential difference between two points is measured by using a device called voltmeter.

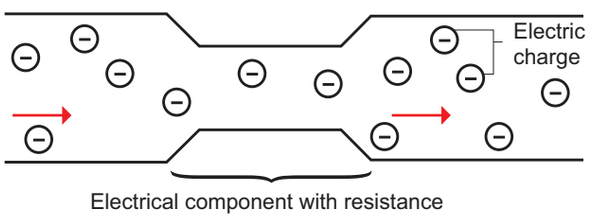
Water at the top of the waterfall has more potential energy



Water near the base of the waterfall has less potential energy

The electric current flow from the higher potential level to the lower potential level is just like the water flow.

2.2.2. Electrical conductivity and Resistivity



Resistance (R)

An electrical component resists or hinders the flow of electric charges, when it is connected in a circuit. In a circuit component, the resistance to the flow of charge is similar to how a narrow channel resists the flow of water.

The higher the resistance in a component, the higher the potential difference needed to move electric charge through the component. We can express resistance as a ratio.

Resistance of a component is the ratio of the potential difference across it to the current flowing through it. $R = \frac{V}{I}$

The SI unit of resistance is ohm

Greater the ratio of V to I, the greater is the resistance

Electrical conductivity (σ)

Electrical conductivity or specific conductance is the measure of a material's ability to conduct an electric current. It is commonly represented by the Greek letter σ (sigma) The SI Unit of electrical conductivity is Siemens/meter(S/m)

Electrical resistivity (ρ)

Electrical resistivity (also known as specific electrical resistance, or volume resistivity) is a fundamental property of a material that quantifies how strongly that material opposes the flow of electric current. The SI unit of electrical resistivity is the ohm-metre (Ω.m).

Material	Resistivity (ρ) (Ω m) at 20°C	Conductivity (σ) (S/m) at 20°C
Silver	1.59×10^{-8}	6.30×10^{7}
Copper	1.68×10^{-8}	5.98×10^{7}
Annealed Copper	1.72×10^{-8}	5.80×10^{7}
Aluminum	2.82×10^{-8}	3.5×10^{7}

2.2.3. Analogy of Electric Current with Water Flow

An electric current is a flow of electrons through a conductor (like a copper wire). We can't see electrons, however, we can imagine the flow of electric current in a wire like the flow of water in a pipe.

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